

SADNAN JONAYET SAJID

Aspiring Autonomous Systems Researcher — SLAM · Sensor Fusion · Embedded AI

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Year 3 of 4, B.Sc. EEE · Team Lead, 3-Person Robotics Research Group · 1 Submitted — Under Review · Seeking Fully Funded MSc / RA, 2027–28

RESEARCH STATEMENT

My path into robotics research did not begin in a classroom. It began in industrial substations and on a family poultry farm, where two years spent keeping generators running and a small business alive through a pandemic taught me that engineering is, first, about delivering reliably under constraint. That instinct now anchors my research in autonomous navigation: leading a three-member team building a fully autonomous mobile robot from raw LiDAR and RGB-D data to real-time navigation decisions on NVIDIA Jetson Nano hardware. I am seeking a fully funded MSc or Research Assistantship to turn this trajectory into rigorous, published research on SLAM, multi-sensor fusion, and field robotics.

EDUCATION

University of Asia Pacific, Dhaka, Bangladesh Jan 2024 – Dec 2027 (Expected)

B.Sc. in Electrical & Electronic Engineering

- Coursework focus: Embedded Systems, Signal Processing, Control Systems, and Applied AI in Engineering Contexts.
- Conducting independent, team-led robotics research in parallel with coursework since Year 2.

RESEARCH EXPERIENCE

Team Leader & Systems Architect — Ultron UV 2024 – Present

University of Asia Pacific · 3-member research team

- Objective:** design and build a fully autonomous mobile robot capable of real-time mapping and navigation on low-cost, resource-constrained edge hardware.
- Lead a three-member team across the full research pipeline: hardware integration (LiDAR, Kinect RGB-D, Arduino Mega, NVIDIA Jetson Nano) and software architecture (ROS2, SLAM Toolbox, Nav2).
- Implemented LiDAR + RGB-D multi-sensor fusion to improve localization robustness in cluttered, dynamic indoor environments.
- Architected a distributed control system over a Zenoh router with a Tailscale mesh VPN for remote monitoring, telemetry, and teleoperation during field testing.
- Status:** hardware integration phase; findings forming the basis of a manuscript currently in preparation (see Publications).

Applied AI Systems (Selected) 2024 – Present

Independent projects exploring agentic AI, RAG pipelines, and automation — methodological range beyond the core robotics research

- RoboWatch** — automated information-retrieval pipeline that filters 34 RSS sources weekly using Gemini 2.5 Flash, surfacing scholarships, deadlines, and competitions.
- Academic AI Automation** — self-triggering pipeline combining fine-tuning and Gemini 2.5 Flash to synthesize structured engineering notes from course materials.
- AI Team Dashboard** — live research-team monitoring system with an embedded Groq-powered AI assistant for implementation support.

PUBLICATIONS & MANUSCRIPTS IN PREPARATION

Sajid, S. J., et al. “Autonomous Mobile Robot Navigation via ROS2-Based SLAM and AI-Enhanced Decision-Making” (working title). University of Asia Pacific. Submitted — Under Review; targeting an IEEE / international robotics conference or journal. Investigates real-time 2D mapping (SLAM Toolbox) and Nav2-based path planning with LiDAR + RGB-D sensor fusion on Jetson Nano edge hardware, with evaluation focused on map consistency under sensor noise and navigation performance in dynamic environments.

RESEARCH INTERESTS

- **Simultaneous Localization & Mapping (SLAM)** — loop closure and map consistency under sensor noise.
- **Multi-Sensor Fusion** — combining LiDAR, IMU, and RGB-D data for robust state estimation.
- **Edge AI for Autonomous Systems** — real-time perception (TensorRT, YOLOv11) under strict compute and power budgets.
- **AI-Integrated Automation & Agentic Pipelines** — applying retrieval-augmented generation and LLM agents to research and engineering workflows.

TECHNICAL & RESEARCH SKILLS

Robotics Middleware: ROS2, Nav2, SLAM Toolbox, Zenoh, Gazebo, TF2, RViz2, micro-ROS

Hardware & Embedded: NVIDIA Jetson Nano, Arduino Mega, Raspberry Pi, LiDAR, Kinect RGB-D, C/C++, TensorRT, I²C/SPI/UART

AI, ML & Data: Python, PyTorch, OpenCV, YOLOv11, LangChain, , RAG Pipelines, ChromaDB

Research Tools: Git/GitHub, Docker, Linux/Ubuntu, LaTeX/Overleaf

HONORS, AWARDS & COMPETITIVE ACHIEVEMENTS

- Champion — IEEE WiE BD Summit 2024, Robotics for Climate Change
- 5th Place — IEEE Region 10 Robotics Competition
- 11th Place — 6th Kibo Robot Programming Challenge (JAXA-affiliated)

LEADERSHIP & AFFILIATIONS

- **Team Leader**, Team Supersonic UAP, University of Asia Pacific (2024–Present).
- **Webmaster**, IEEE UAP SB WIE Affinity Group, IEEE UAP Student Branch.
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- IEEE Student Member.

PRIOR PROFESSIONAL EXPERIENCE

Before returning to full-time study, I spent two years in industrial and operational roles that grounded my engineering instincts in real-world delivery constraints — experience that now directly informs how I plan and execute long-term research projects.

- **Project Manager, Golden Power Engineering** (2023–2025) — progressed through four roles in two years while managing industrial power infrastructure projects (3-phase generators, substations, high-voltage distribution) for commercial and industrial clients.
- **Operations Lead, Family Agribusiness** (2020–2022) — sole operational responsibility for a poultry and feed business through the COVID-19 disruption, including supply chain, procurement, and distribution.

LANGUAGES

English: Professional Working Proficiency

Bengali: Native / Bilingual Proficiency

REFERENCES

Available upon request.